

Tinghui Li

Gender: Female

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EDUCATION

University of Sydney

Doctor of Philosophy

Sydney, Australia

10/2023 – 03/2027

- **Supervisors:** Dr. Zhanna Sarsenbayeva, A/Prof. Anusha Withana, Prof. Eduardo Velloso
- **Research Area:** Human-Computer Interaction, Extended Reality, Accessibility
- **Project Topic:** Understanding and Modelling the Effects of Situational Impairments in Extended Reality
- **Scholarships:** Full Tuition Fee Scholarship, Full Stipend Scholarship

University of Sydney

Bachelor of Advanced Studies (Honours)

Sydney, Australia

09/2022 – 09/2023

- **Major:** Computer Science (First-Class Honours)
- **Supervisor:** A/Prof. Anusha Withana
- **Project Topic:** Physics Inspired Time-Agnostic Graph Neural Network for Finger Motion Prediction

University of Sydney

Bachelor of Science

Sydney, Australia

02/2020 – 09/2022

- **Majors:** Data Science, Biochemistry and Molecular Biology
- **Courses:** Human-in-the-Loop Data Analytics (91/100); Pervasive Computing (91/100); Introduction to Programming (89/100); Visual Analytics (88/100); Usability Engineering (82/100).

EXPERIENCE

School of Computer Science, University of Sydney

Casual Academic – Computer Science

Sydney, Australia

06/2023 – Present

- COMP4427/5427 Usability Engineering Guest Lecturer
- COMP4427/5427 Usability Engineering Teaching Assistant and Tutor
- INFO3315 Human-Computer Interaction Teaching Assistant and Tutor
- COMP5703/DATA5703/CSEC5703 Group-Based Capstone Project Tutor and Client

Graduate School of Media Design, Keio University

Visiting Researcher

Tokyo, Japan

04/2025 – 05/2025

- **Supervisor:** Prof. Kouta Minamizawa
- **Project Topic:** Estimating the Effects of Ambient Noise on Mixed Reality Interaction

Bosch (China) Investment Co., Ltd.

Data Scientist Intern

Shanghai, China

11/2021 – 01/2022

- Analysed the abnormal vehicles data, including finding the abnormal cells, plotting current and voltage graphs;
- Participated in the production and improvement of the PowerBI dashboard.

University College London

Exchange Summer School

London, England

07/2021 – 08/2021

- **Course:** Health Data Science and Data Analytics in Healthcare.

Shanghai Jiao Tong University

Exchange Summer School

Shanghai, China

12/2020 – 01/2021

- **Courses:** Algorithms and Analysis (89/100); Statistics of Stochastic Processes (89/100).

PUBLICATIONS

- [CHI'25 Honorable Mention] **Tinghui Li**, Eduardo Velloso, Anusha Withana, and Zhanna Sarsenbayeva (2025). Estimating the Effects of Encumbrance and Walking on Mixed Reality Interaction. In Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems 2025 (CHI'25, **CORE A***).
- [UIST'25] **Tinghui Li**, Eduardo Velloso, Anusha Withana, and Zhanna Sarsenbayeva. Weight-Induced Consumed Endurance (WICE): A Model to Quantify Arm Fatigue when Carrying Objects across Different Arm Locations. In Proceedings of the ACM Symposium on User Interface Software and Technology 2025 (UIST'25, **CORE A***).
- [arXiv] **Tinghui Li**, Pamuditha Somarathne, Zhanna Sarsenbayeva, and Anusha Withana. TA-GNN: Physics Inspired Time-Agnostic Graph Neural Network for Finger Motion Prediction. arXiv Preprints 2025.
- [Under Review] **Tinghui Li**, Danyang Peng, Eduardo Velloso, Anusha Withana, Kouta Minamizawa, and Zhanna Sarsenbayeva. Estimating the Effects of Ambient Noise on Mixed Reality Interaction.
- [Under Review] **Tinghui Li**, Brandon Victor Syiem, Niels van Berkel, Juno Kim, Anusha Withana, and Zhanna Sarsenbayeva. Trends and Directions of Extended Reality Research in HCI: A Co-Word Analysis.
- [Under Review] **Tinghui Li**, Pamuditha Somarathne, Zhanna Sarsenbayeva, and Anusha Withana. Never-Lagging: Enhancing Virtual Reality Finger Tracking with a Physics-Inspired Time-Agnostic Graph Neural Network.
- [Under Review] **Tinghui Li**, Guangzheng Zhang, and Zhanna Sarsenbayeva. SOBO: An AI-Driven Small-Object Boundary Detection Framework for Optimizing VR Safety in Confined Spaces.
- [Under Review] **Tinghui Li**, Lefan Lai, Yingqi Wang, and Zhanna Sarsenbayeva. Exploring Virtual Reality Interaction in Confined Spaces: A Third-Person Perspective Solution.
- [Under Review] Ruilin Liu, **Tinghui Li**, and Zhanna Sarsenbayeva. Understanding and Addressing Ambient Illumination as a Situational Impairment for Digital Devices.
- [Under Review] Ruilin Liu, **Tinghui Li**, and Zhanna Sarsenbayeva. Estimating the Effects of Ambient Light on Mixed Reality Interaction.
- [Under Review] Danyang Peng, Zicheng Xia, **Tinghui Li**, Yixin Wang, Mark Armstrong, Kinga Skiers, Anish Kundu, Kouta Minamizawa, and Yun Suen Pai. Bioresponsive Avatars: Perceiving Emotions Through Virtual Avatar Representation in Empathic Social VR.
- [Ongoing] **Tinghui Li**, Juno Kim, Eduardo Velloso, Anusha Withana, and Zhanna Sarsenbayeva. Hierarchical Quantification of Factors Underlying the Effects of Walking on Mixed Reality Interaction.
- [Ongoing] Zheyuan Kuang, **Tinghui Li**, Weiwei Jiang, Sven Mayer, Flora Salim, Benjamin Tag, Anusha Withana, Zhanna Sarsenbayeva. Understanding the Effects of Interactivity on Emotional Experiences in VR.
- [Ongoing] Lefan Lai, **Tinghui Li**, Zhanna Sarsenbayeva, and Brandon Syiem. Searching Through Complex Worlds: Visual Search and Spatial Regularity Memory in Mixed Reality.

AWARDS

- 2025 Deputy Vice-Chancellor Research Impact Proof-of-Concept Fund (A\$79,874)
- 2025 Faculty HDR Grant (A\$800)
- 2025 University of Sydney CS HDR Research Support Fund (A\$5,000)
- 2023 – 2027 University of Sydney Faculty of Engineering Research Stipend Scholarship (A\$40,109 p.a.)
- 2023 – 2027 University of Sydney International Tuition Fee Scholarship
- 2023 University of Sydney Engineering Vacation Research Internship Program Scholarship (A\$3,000)
- 2023 University of Sydney Undergraduate High Honour Roll
- 2022 University of Sydney Undergraduate Coursework and Top 3rd Year High Honour Roll
- 2021 University of Sydney Global Citizenship Award

ACTIVITIES

- **Review Committee Member:** UbiComp/ISWC'25.
- **Reviewer:** CHI'25, ICXR'25, OzCHI'25, OzCHI'24, AHs'24.
- **Student Volunteer:** ISMAR'23, VLDB'22.